

# How to Test the MQTT Broker with TLS/mTLS Enabled

## 1. Start the Broker

Open PowerShell and run:

```
cd C:\custom_secure_mqtt_broker\secure-mqtt-broker
.\start_broker.bat
```

Leave this window open.

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## 2. Open a Second PowerShell Window

Run a secure subscriber using the client certificate:

```
mosquitto_sub -h localhost -p 8883 `
--cafile C:\custom_mqtt_broker\certificates\ca.crt `
--cert C:\custom_mqtt_broker\certificates\client.crt `
--key C:\custom_mqtt_broker\certificates\client.key `
-t sensors/#
```

Leave this window waiting for incoming messages.

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## 3. Open a Third PowerShell Window

Publish a temperature value using the same trusted certificate:

```
mosquitto_pub -h localhost -p 8883 `
--cafile C:\custom_mqtt_broker\certificates\ca.crt `
--cert C:\custom_mqtt_broker\certificates\client.crt `
--key C:\custom_mqtt_broker\certificates\client.key `
-t sensors/temperature `
-m "25.6"
```

Expected result:

The subscriber window should display:

```
25.6
```

This confirms that:

- The TLS handshake succeeds.
- The broker certificate is trusted.
- The client certificate is trusted.
- Secure MQTT communication is established.

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## 4. Test Connection Without a Certificate

Attempt to connect without providing a client certificate:

```
mosquitto_sub -h localhost -p 8883 `
--cafile C:\custom_mqtt_broker\certificates\ca.crt `
-t sensors/#
```

Expected result:

Connection should fail.

Example error:

```
The connection was lost.
```

or

```
Connection Refused
```

This confirms that the broker requires client certificates for authentication.

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## 5. Test Connection with an Invalid Certificate

If available, use a certificate that was not signed by your CA:

```
mosquitto_sub -h localhost -p 8883 `
--cafile C:\custom_mqtt_broker\certificates\ca.crt `
--cert invalid.crt `
--key invalid.key `
-t sensors/#
```

Expected result:

The connection should be rejected during the TLS handshake.

This confirms that only certificates signed by the trusted CA are accepted.

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## 6. Test Broker Certificate Verification

Attempt to connect without the CA certificate:

```
mosquitto_sub -h localhost -p 8883 `
--cert C:\custom_mqtt_broker\certificates\client.crt `
--key C:\custom_mqtt_broker\certificates\client.key `
-t sensors/#
```

Expected result:

The client should refuse the connection because it cannot verify the broker's identity.

This confirms that clients verify the broker certificate before communicating.

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## Expected Outcome

The following security properties should be verified:

- Only encrypted MQTT communications are allowed.
- Clients verify the broker's identity using the CA certificate.
- The broker verifies the client's identity using the client certificate.
- Clients without certificates cannot connect.
- Certificates not signed by the trusted CA are rejected.
- Mutual TLS (mTLS) authentication is enforced successfully.

These tests demonstrate the correct implementation of transport security using TLS and mutual certificate authentication.